

Rithish S

Software Engineering Intern | Distributed Systems, Algorithms & Cloud Services

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EDUCATION

Bannari Amman Institute of Technology
Bachelor of Engineering in Computer Science and Engineering
CGPA: **8.28/10**

Sathyamangalam, Tamil Nadu
Aug. 2024 – May 2028

TECHNICAL SKILLS

Languages: C++, Java, Python, TypeScript, JavaScript, SQL, Go
Computer Science: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Networks, Distributed Systems
Backend & Systems: Node.js, Fastify, Express.js, REST APIs, Redis (Distributed Queues & Locks), BullMQ, Microservices, k6
Databases & Cloud: PostgreSQL, MySQL, MongoDB, SQLite, Pinecone, AWS (EC2, S3), Docker, Linux (POSIX), Git, CI/CD
Certifications: AWS Cloud Practitioner, Oracle Cloud Foundations Associate, Cisco OS & Linux, NPTEL Python (Top 5%)

EXPERIENCE

Web Developer Feb. 2025 – Dec. 2025
Students Special Group *Student Organization*

- Engineered full-stack internal management portals with React, Node.js, and MySQL, implementing normalized relational schemas and role-based workflows for institutional operations.
- Designed and deployed high-throughput REST APIs for a Faculty Dashboard, optimizing SQL queries, indexing, and transactional integrity across high-frequency academic updates.
- Mentored **50+ students** in Data Structures & Algorithms, backend architecture, version control workflows, and software engineering design patterns.

SOFTWARE ENGINEERING PROJECTS

DB Backup CLI (DBVault) | *TypeScript, Node.js, Redis, BullMQ, AWS S3, Docker* [GitHub](#)

- Architected an automated multi-engine backup/restore CLI published on npm and Docker Hub with **120 automated tests** supporting PostgreSQL, MySQL, MongoDB, and SQLite.
- Designed a distributed asynchronous job processing pipeline using Redis and BullMQ, implementing worker concurrency control, queue rate-limiting, and distributed locking.
- Implemented AES-256-GCM authenticated encryption with local key rotation, stream-based compression, and automated cloud offloading via **AWS S3** object storage adapters.

PatentIQ | AI Prior-Art & RAG Engine | *Node.js, Fastify, Python, PostgreSQL, Pinecone, Prisma* [GitHub](#)

- Engineered a semantic prior-art discovery engine utilizing Pinecone vector search and LLM embeddings to retrieve patent claims from high-dimensional vector spaces.
- Developed a RAG pipeline leveraging Qwen2.5 for automated patent claim-overlap analysis, technical novelty scoring, and contextual risk evaluation.
- Architected a modular Fastify backend using Clean Architecture, dependency injection, Prisma ORM, PostgreSQL transactional persistence, and Zod schema validation.

OPEN SOURCE

ESLint | *JavaScript, AST, Static Analysis, IEEE 754, CI/CD* [PR #21218](#)

- Contributed a merged bug fix to ESLint's **no-loss-of-precision** static analysis rule, correctly differentiating true zero literals from non-zero values underflowing IEEE 754 binary64 precision.
- Authored comprehensive unit and regression tests covering numeric underflow, subnormal IEEE 754 boundaries, and signed zero; successfully passed **43 CI validation checks**.

ACHIEVEMENTS

- 1200+ Algorithmic Problems Solved** across LeetCode, CodeChef, and GeeksforGeeks, demonstrating advanced proficiency in Dynamic Programming, Graph Theory, Trees, and Greedy Algorithms.
- IEEE DevSpark Hackathon Winner** – Secured **1st place among 120+ teams** in Full Stack Development by architecting Plantera, an AI-powered environmental monitoring platform.
- BIT Hackathon Winner** – Secured **1st place among 200+ teams** by engineering a full-stack Leave Management System under strict 24-hour hackathon constraints.
- NPTEL Elite + Topper (Top 5%)** – Scored 95% in The Joy of Computing using Python (IIT Madras); certified in AWS Cloud Practitioner and Oracle Cloud Foundations Associate.